

Question 1

- a) Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think, learn, and make decisions like humans. To develop an AI system, four different approaches are implemented: Acting Humanly, Thinking Humanly, Thinking Rationally, and Acting Rationally. Briefly explain the **FOUR (4)** approaches. (8 marks)

Acting Humanly

- Known as the Turing Test approach.
- AI systems developed with this approach are capable of acting like human.

Thinking Humanly

- Known as the Cognitive Modelling approach.
- AI systems developed with this approach are capable of thinking like human.

Thinking Rationally

- Known as the Laws of Thought approach.
- AI systems developed with this approach are capable of thinking rationally.

Acting Rationally

- Known as the Rational Agent approach.
- AI systems developed with this approach are capable of acting rationally.

- b) Many modern technologies are related to or inspired by the idea behind the Turing Test. One example is **Meta AI**, an artificial intelligence assistant created by Meta, the company that owns Facebook, Instagram, WhatsApp, and Messenger. Meta AI is built into these apps to help users with different tasks. It can answer questions, give suggestions, help you chat, create images or animations, and assist with personalising or creating content. In simple words, Meta AI is like a smart helper inside Meta's apps that makes everyday tasks easier.

- (i) Identify how the *total Turing Test* is different from the *Turing Test*. (2 marks)

The total Turing test has two additional requirements added to create a stronger benchmark of intelligence:

- Perceptual ability (OR to derive visual information from its percepts)
- Object manipulation ability (OR to manipulate objects and move about)

- (ii) Identify whether Meta AI can pass the *Turing Test*. Justify your answer by focusing on the capabilities of Meta AI. (1 + 2 marks)

Yes (partially)

- Can communicate in human languages (Natural Language Processing)
- Can adapt to new circumstances and handle uncertainties (Machine Learning)

- (iii) Identify whether Meta AI can pass the *total Turing Test*. Justify your answer by focusing on the capabilities of Meta AI. (1 + 2 marks)

No. The reason is that Meta AI is not capable of either one or both of the following:

- Robotics (OR it can manipulate objects and move about)

- (iv) Which one is more relevant to the definition of AI, the *Turing Test* or the *total Turing Test*? Justify your answer. (1 + 2 marks)

The total Turing test. The reason is that human can derive information through vision and manipulating objects and move about.

- c) Define agent in terms of AI. (2 marks)

An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.

- d) At any given time, rationality depends on four criteria to develop a rational agent. Briefly explain the **FOUR** **(4)** criteria.
(4 marks)

- The performance measure that defines the criterion of success.
- The agent's prior knowledge of the environment.
- The actions that the agent can perform.
- The agent's percept sequence to date.

Question 2

A company developing smart traffic control systems aims to optimise the timing of traffic lights across a busy city to reduce congestion and improve traffic flow. Due to the dynamic nature of vehicle movements and unpredictable traffic patterns, traditional algorithms struggle to maintain efficiency during peak hours. To overcome this, the company implements an evolutionary algorithm (EA) to optimise traffic signal timings. The algorithm evolves different timing strategies and evaluates them based on average vehicle waiting time, traffic flow rate, and fuel efficiency.

- a) Analyse how an evolutionary algorithm can help improve the traffic light timing system. (2 marks)

- An evolutionary algorithm improves traffic light timing by iteratively generating, evaluating, and evolving multiple timing strategies based on fitness measures such as reduced waiting time and improved traffic flow.
- It enables the system to adapt dynamically to changing traffic conditions, achieving greater efficiency than traditional static methods.

- b) Identify and briefly explain the **FIVE (5)** phases of an evolutionary algorithm involved in optimising the traffic light timing. (5 + 10 marks)

Beginner-friendly version

Initialization → Randomly create many possible traffic-light timing plans.

Evaluation → Test each plan in a traffic simulation and calculate its fitness based on waiting time, traffic flow, fuel consumption, etc.

Selection → Choose the better-performing plans to become parents.

Crossover → Combine parts of two parents to create new timing plans (offspring).

Mutation → Make small random changes to the offspring to maintain diversity and avoid getting stuck at a poor solution.

Repeat → Evaluate the new population and repeat the process until a satisfactory solution is found.

Initialization

- A population of potential traffic signal timing strategies is randomly generated.
- Each individual (chromosome) represents a possible configuration of green, yellow, and red light durations for multiple intersections.
 - (Green = 30s, Yellow = 5s, Red = 40s) - 5 min
 - (Green = 20s, Yellow = 15s, Red = 40s) - 15 min
 - (Green = 40s, Yellow = 5s, Red = 30s) - 50 min

Evaluation (Fitness Function)

- Each strategy is evaluated using a simulation of city traffic flow.
- Metrics such as average vehicle waiting time, traffic throughput, and fuel consumption are measured to determine the fitness score of each solution.

Selection

- The best-performing strategies are selected based on their fitness values.
- Methods such as tournament selection or roulette wheel selection may be used to ensure that more efficient timing plans have a higher chance of being chosen for reproduction.
 - Parent A: Green = 30s, Yellow = 5s, Red = 40s - 5min
 - Parent B: Green = 50s, Yellow = 5s, Red = 20s - 3min

Crossover (Recombination)

- Selected strategies are paired and combined to produce new offspring.
- For example, timing plans from two intersections can be blended to create a new, potentially more efficient hybrid timing configuration.
 - Child A: Green = 30s, Yellow = 5s, Red = 20s
 - Child B: Green = 50s, Yellow = 5s, Red = 40s

Mutation

- Small random adjustments are introduced to offspring solutions, such as slightly increasing or decreasing the green light duration.
- Mutation ensures diversity in the population and prevents the algorithm from converging prematurely to a local optimum.
 - Mutated Child A: Green = 33s, Yellow = 5s, Red = 20s - 5 min

- c) Identify and briefly explain **TWO (2)** stopping criteria for the evolutionary algorithm that ensure optimal performance without excessive computation time. Justify your recommendation.

(8 marks)

Maximum Number of Generations

- The algorithm stops after a predefined number of generations (e.g., 200). This prevents excessive computation while ensuring enough iterations for convergence.
- Justification: This criterion provides a clear limit on processing time and is suitable for real-time systems like traffic control that require periodic updates.

Fitness Convergence Threshold

- The algorithm stops when the improvement in the best fitness score between consecutive generations falls below a small threshold (e.g., less than 0.01% improvement over 10 generations).
- Justification: This ensures the algorithm halts once it reaches a stable and efficient traffic timing plan, avoiding unnecessary computation when no further significant improvement occurs.

Question 3

- a) In AI, game tree search refers to examining all possible moves and their outcomes in a game to determine the optimal decision. Based on the game tree shown in **Figure 1**, answer the following questions.

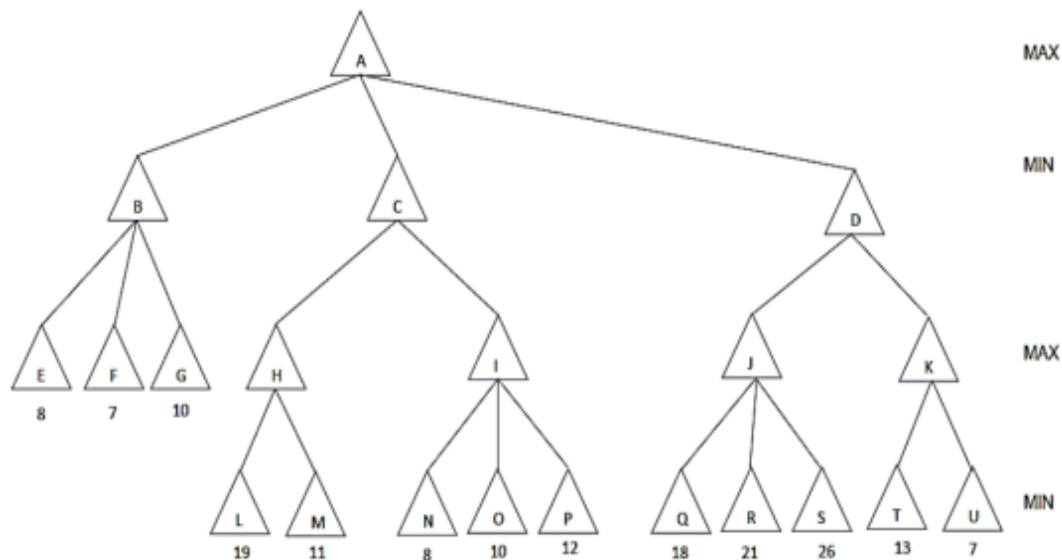


Figure 1. A partial game tree that illustrates the gameplay of a chess game

- (i) The game represented by the game tree in **Figure 1** demonstrates that the environment involves multiple agents. Identify and briefly explain the specific type of multi-agent environment depicted. (2 marks)

- Competitive
- In this environment, agents compete against each other.

- (ii) Suppose the *minimax algorithm* is used to explore the game tree in search of the optimal move, compute the *minimax value* for nodes A, B, C, D, H, I, J, and K. (4 marks)

- A = 13
- B = 7
- C = 12

- $D = 13$
- $H = 19$
- $I = 12$
- $J = 26$
- $K = 13$

- (iii) Identify the best move for the MAX player at node A. Justify your answer. (2 marks)

A-D, because this action results in a state with the highest minimax value.

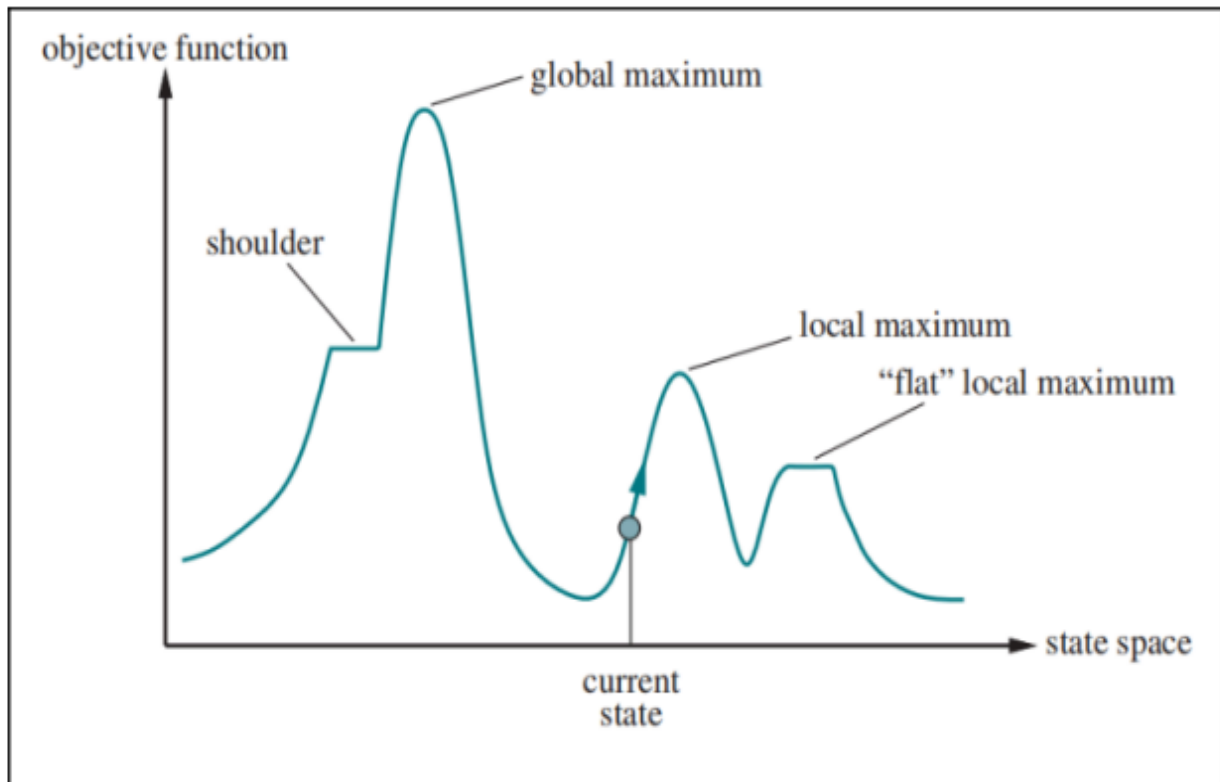
- b) *Hill climbing* is an optimisation algorithm used to identify the best possible solution for a problem. It is commonly applied in AI and machine learning for local search optimisation.
- (i) *Hill climbing* is also known as a greedy approach or greedy local search. Briefly explain why it is referred to in this way. (2 marks)

Only consider a good neighbouring state without thinking ahead.

- (ii) Give **TWO (2)** strengths of *Hill Climbing*. (2 marks)

- Have very low memory consumption.
- Often can find reasonable solutions in large or infinite state spaces.

- (iii) Draw and label a state-space landscape in which its elevation corresponds to an objective function. (7 marks)



- (iv) Hill climbing search often gets stuck at a sub-optimal solution. Identify and analyse **TWO (2)** causes for this. (2 + 4 marks)

Local maxima

- A peak that is higher than each of its neighbouring states but lower than the global maximum.
- Once arrived at the local maxima, it will be stuck with nowhere else to go.



Ridges

- A sequence of local maxima that is very difficult for greedy algorithms to navigate.
- From each local maximum, all the available actions point downhill.



Plateaus

- A flat area of the state-space landscape, a flat local maximum, and a shoulder.
- Search can get lost wandering on the plateau.



Question 4

The IT Department at TAR UMT has developed an AI-driven campus Wi-Fi optimization system designed to identify the most efficient connection path from Access Point A to Access Point I. **Figure 2** illustrates a directed network graph consisting of nine access points, where the values represent the signal transmission costs between each point. **Table 1** shows the heuristic cost of each access point relative to Access Point I, assisting the system in estimating the optimal data transmission route.

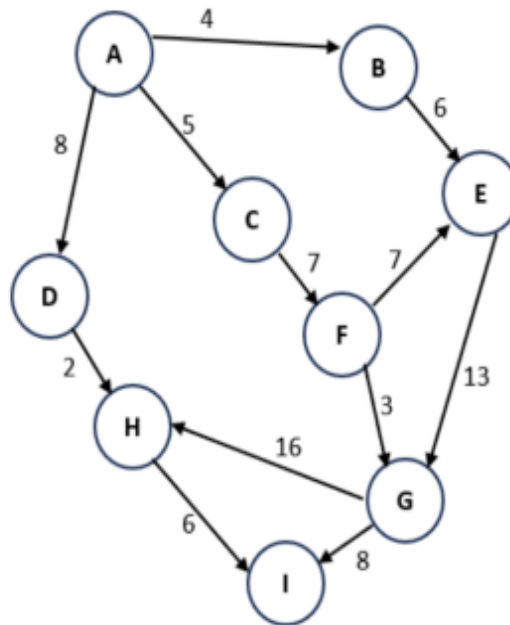


Figure 2. State space

Table 1: Heuristic Cost of Each Access Point

Access Point	A	B	C	D	E	F	G	H	I
Heuristic Cost	13	17	6	9	19	7	6	9	0

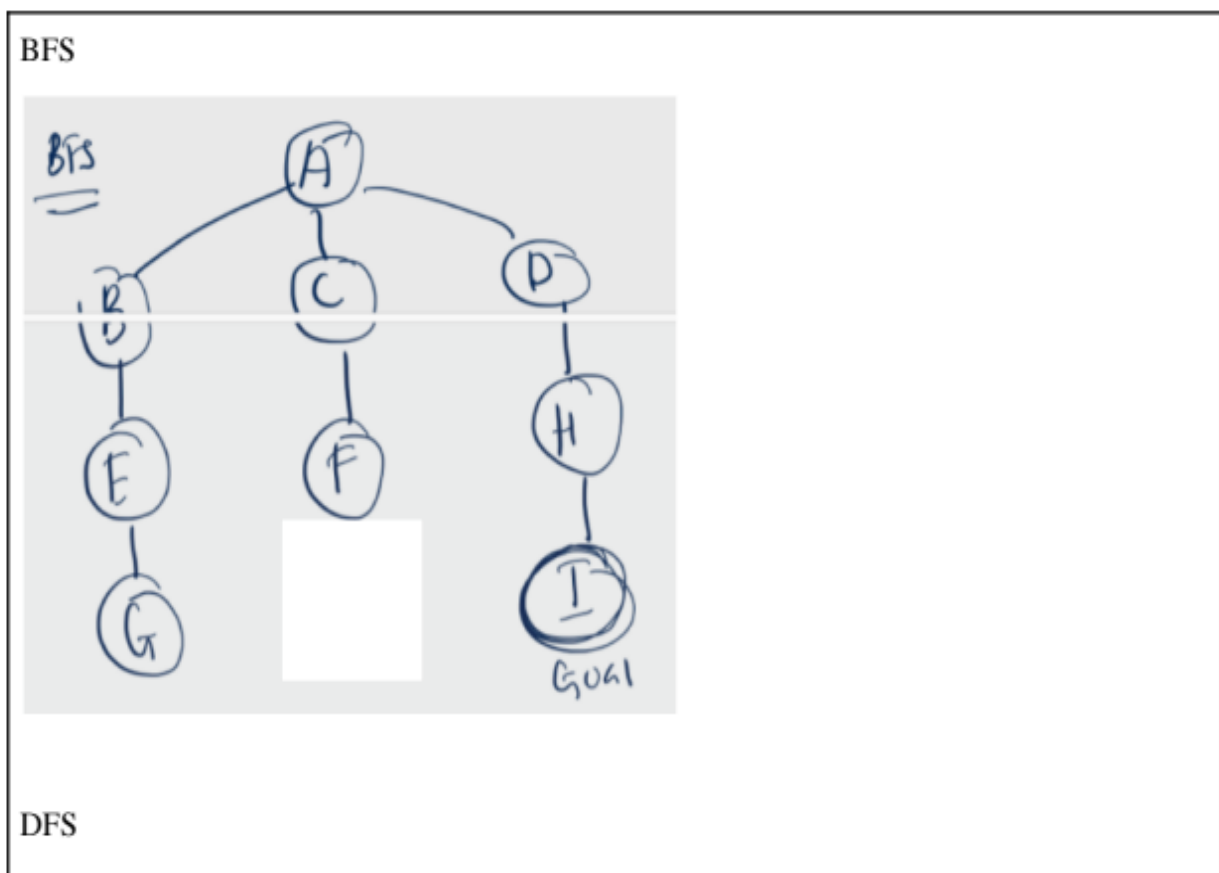
- a) Formulate the state space, initial state, goal state, actions, transition model, and the cost function for the problem above. (6 marks)

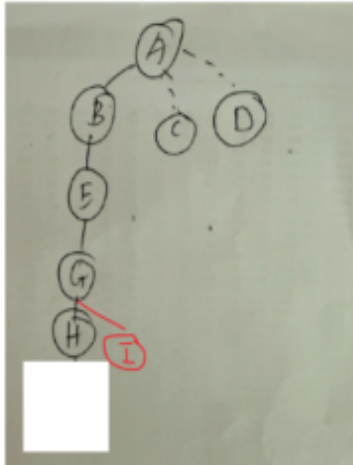
States space	The set of nodes in the search graph (A, B, C, ..., H)
Initial state	Starting node (A)
Goal state	Target node (I)
Action	Possible moves from one node to another

	based on the graph's edges
Transition model	Defines how each action moves from one node to another and updates the path cost.
Cost function	The evaluation function for A* is $f(n) = g(n) + h(n)$, where: • $g(n)$ = actual cost from the start node to node n • $h(n)$ = estimated heuristic cost from n to the goal

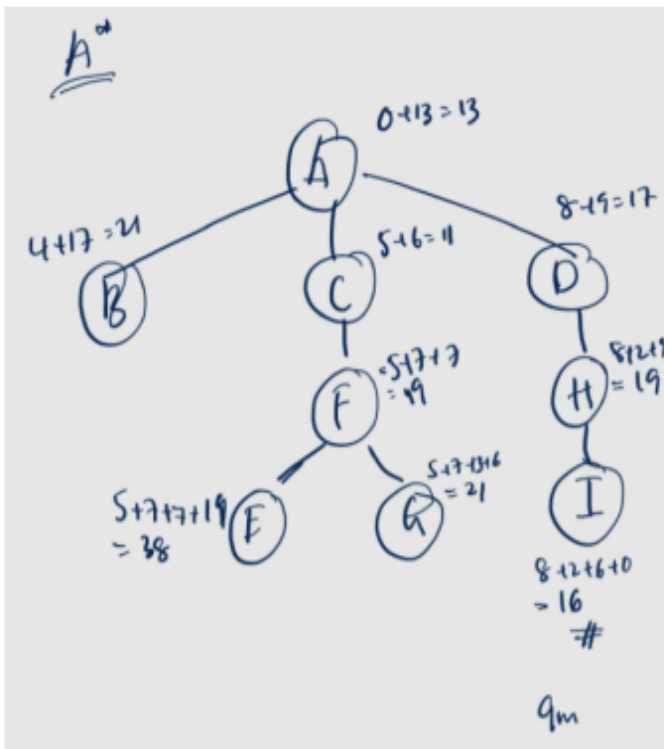
- b) Based on the information provided in **Figure 2** and **Table 1**, illustrate the resulting search tree for *Breadth-First Search*, *Depth-First Search*, and *A* Search* in alphabetical order.

(16 marks)





A*



Expanded Node	Frontier
	A13
A13	B21, C11, D17
C11	B21, D17, F19
D17	B21, F19, H19

F19	B21, H19, E38, G21
H19	B21, E38, G21, I16

- c) Identify the solution returned by *Breadth-First Search*, *Depth-First Search*, and *A* Search* in **Question 4 b)**. (3 marks)

- BFS: A,D, H, I (Total path cost = 16)
- DFS:A,B,E, G, H, I
- A* : A,D,H,I